

BOARD QUESTION PAPER : JULY 2025

MATHEMATICS AND STATISTICS

Time: 3 Hrs.

Max. Marks: 80

General Instructions:

- (i) All questions are compulsory.
- (ii) There are six questions divided into two sections.
- (iii) Write answers of Section-I and Section-II in the same answer book.
- (iv) Use of logarithmic tables is allowed. Use of calculator is not allowed.
- (v) For L.P.P. and Time Series graph paper is not necessary. Only rough sketch of graph is expected.
- (vi) Start answer to each question on a new page.
- (vii) For each objective type of question (i.e. Q.1 and Q.4) only the first attempt will be considered for evaluation.

Section – I

Q.1. (A) Select and write the correct answer of the following multiple choice type of questions (1 mark each): (6) [12]

- i. Matrix $B = \begin{bmatrix} 0 & 3 & 1 \\ -3 & 0 & -4 \\ P & 4 & 0 \end{bmatrix}$ is skew-symmetric then value of P is
- (a) 1 (b) -1 (c) 0 (d) -3
- ii. If $y = e^{\log x}$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$.
- (a) $\frac{1}{x}$ (b) $\frac{1}{2}$ (c) $\frac{e^{\log x}}{x}$ (d) 0
- iii. $\int (1-x)^{-3} dx = \underline{\hspace{2cm}}$.
- (a) $\frac{1}{2} (1-x)^{-2} + c$ (b) $\frac{1}{2} (1+x)^{-2} + c$
(c) $\frac{1}{2} (1-x)^{-2} + \frac{x}{2} + c$ (d) $\frac{1}{2} (1-x)^{-2} - \frac{x}{2} + c$
- iv. If $\int_0^a 3x^2 dx = 8$ then $a = \underline{\hspace{2cm}}$.
- (a) 0 (b) 2 (c) 8 (d) $\frac{4}{3}$
- v. Area of the region bounded by the curve $y = x^2$, the X-axis and the lines $x = 1$ and $x = 3$ is $\underline{\hspace{2cm}}$.
- (a) $\frac{3}{26}$ sq. units (b) 3 sq. units (c) 26 sq. units (d) $\frac{26}{3}$ sq. units
- vi. The order and degree of $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 = a^x$ are respectively $\underline{\hspace{2cm}}$
- (a) 1, 1 (b) 1, 2 (c) 2, 1 (d) 2, 2
- (B) State whether the following statements are true or false (1 mark each):** (3)
- i. Dual of $(p \wedge \sim q) \vee t$ is $(p \vee \sim q) \vee c$
- ii. For $\int \frac{x-1}{(x-1)^3} e^x dx = e^x \cdot f(x) + c$, where $f(x) = (x+1)^2$
- iii. The integrating factor (I.F.) of $\frac{dy}{dx} + y = e^{-x}$ is e^x

(C) Fill in the following blanks (1 mark each): (3)

- Negation of "Some men are animals" is _____.
- If the average revenue is 45 and elasticity of demand is 3 then marginal revenue is _____.
- To find the value of $\int \frac{10x^9 + 10^x \cdot \log 10}{10^x + x^{10}} dx$, the proper substitution is _____.

Q.2. (A) Attempt any TWO of the following questions (3 marks each): (6) [14]

- Determine whether the following statement pattern is tautology, contradiction or contingency.
 $[(p \wedge q) \vee (\sim p)] \vee [p \wedge (\sim q)]$
- Find $\frac{dy}{dx}$ if $y = (\log x)^x + x^5$
- Find the area of the region bounded by the parabola $y^2 = 4x$ and line $x = 3$.

(B) Attempt any TWO of the following questions (4 marks each): (8)

- Find MPC, MPS, APC and APS, if the expenditure E_c of a person with income I is given as $E_c = (0.0003) I^2 + (0.075) I$ when $I = 1000$.
- Solve the following differential equation:
 $x^2 \cdot y dx - (x^3 + y^3) \cdot dy = 0$
- Express the following equations in matrix form and solve them by method reduction:
 $x + 2y + z = 8$
 $2x + 3y - z = 11$
 $3x - y - 2z = 5$

Q.3. (A) Attempt any TWO of the following questions (3 marks each): (6) [14]

- Write the converse, inverse and contrapositive of the following statement.
'If a man is bachelor, then he is happy.'
- Find the inverse of $\begin{bmatrix} 3 & 1 & 5 \\ 2 & 7 & 8 \\ 1 & 2 & 5 \end{bmatrix}$ by adjoint method.
- If $x^5 \cdot y^7 = (x + y)^{12}$ then show that $\frac{dy}{dx} = \frac{y}{x}$.

(B) Attempt any ONE of the following questions (4 marks each): (4)

- Evaluate: $\int x^2 \cdot e^{3x} dx$
- Evaluate: $\int_1^4 \frac{\sqrt[3]{x+6}}{\sqrt[3]{x+6} + \sqrt[3]{11-x}} dx$

(C) Attempt any ONE of the following questions (Activity) (4 marks each): (4)

- Find the values of x , such that $f(x)$ is increasing function
 $f(x) = 2x^3 - 15x^2 - 144x - 7$
Solution:
Given: $f(x) = 2x^3 - 15x^2 - 144x - 7$
 $\therefore f'(x) = 6x^2 - 30x - 144$
Now, $f'(x) > 0$, as $f(x)$ is increasing
 $\therefore 6x^2 - 30x - 144 > 0$
 $\therefore x^2 - 5x - 24 > 0$
 $\therefore (x - 8)(x + 3) > 0$
Case (I) $x - 8 > 0$ and $x + 3 > 0$
 $x > 8$ and $x > -3$
 $\therefore x > \boxed{}$
Case (II) $x - 8 < 0$ and $x + 3 < 0$
 $x < 8$ and $x < -3$
 $\therefore x < \boxed{}$
 $\therefore f(x) = 2x^3 - 15x^2 - 144x - 7$ is increasing if and only if $x \in (-\infty, \boxed{})$ or $x \in (\boxed{}, \infty)$

- ii. Solve the following differential equation, hence find the particular solution when $x = 0, y = 1$

$$y^3 - \frac{dy}{dx} = x \frac{dy}{dx}$$

Solution:

$$y^3 = x \frac{dy}{dx} + \frac{dy}{dx}$$

$$\therefore y^3 = (x + 1) \frac{dy}{dx}$$

$$\therefore (x + 1) dy = y^3 dx$$

Separating the variables, we get

$$\frac{1}{y^3} dy = \frac{1}{x+1} dx$$

Now, integrating, we get

$$\therefore \int \frac{1}{y^3} dy = \int \frac{1}{x+1} dx$$

$$\therefore -\frac{1}{2y^2} = \boxed{} + c \quad \dots(i)$$

which is required general solution

put $x = 0$ and $y = 1$ in (i)

$$-\frac{1}{2(1)^2} = \log |0 + 1| + c$$

$$\therefore \boxed{} = c$$

$$\therefore -\frac{1}{2y^2} = \boxed{} - \frac{1}{2}$$

is the particular solution.

Section – II

Q.4. (A) Select and write the correct answer of the following multiple choice type of questions (1 mark each):

(6) [12]

- The sum due is also called as ____
 (a) True discount (b) Face value
 (c) Present value (d) Cash value
- Following are different types of insurance
 (I) Life insurance (II) Health insurance
 (III) Liability insurance
 (a) Only II (b) Only III (c) Only I (d) All the three
- $|b_{xy} + b_{yx}| \geq$ ____
 (a) $2|r|$ (b) $2r$ (c) r (d) $|r|$
- If $P_{01}(L) = 120.4$ and $P_{01}(P) = 130.6$ then $P_{01}(D - B)$ is ____
 (a) 25.1 (b) 60.2 (c) 125.5 (d) 65.3
- $F(x)$ is c.d.f. of discrete r.v. X whose distribution is

X_i	-2	-1	0	1	2
P_i	0.2	0.3	0.15	0.25	0.1

then $F(-3) =$ ____

- (a) 1 (b) 0.2 (c) 0.15 (d) 0

- vi. Given p.d.f. of a continuous r.v. X as

$$f(x) = \frac{x^2}{3}, \text{ for } -1 < x < 2$$

= 0, otherwise

then $F(1) =$ ____

- (a) $\frac{3}{9}$ (b) $\frac{4}{9}$ (c) $\frac{1}{9}$ (d) $\frac{2}{9}$

(B) State whether the following statements are true or false (1 mark each): (3)

- The banker's discount is also called as commercial discount.
- The optimum value of the objective function of L.P.P. occurs at the centre of the feasible region.
- If $E(X) > \text{Var}(X)$ then X follows Binomial distribution.

(C) Fill in the following blanks (1 mark each): (3)

- The region represented by the inequality $y \leq 0$ lies in _____ quadrants.
- The time required for printing of four books A, B, C and D is 5, 8, 10 and 7 hours. While its data entry requires 7, 4, 3 and 6 hours respectively. The sequence that minimizes total elapsed time is _____.
- If X has Poisson distribution with parameter m and $P(X = 3) = P(X = 4)$ then $m =$ _____.

Q.5. (A) Attempt any TWO of the following questions (3 marks each): (6) [14]

- A person wants to create a fund of ₹ 6, 96, 150 after 4 years at the time of his retirement. He decided to invest a fixed amount at the end of every year in a bank that offers him interest of 10% p.a. compounded annually. What amount should he invest every year? [Given $(1.1)^4 = 1.4641$]

- Following are given information about advertising expenditure and sales.

	Advertisement expenditure (₹ in lakh) (X)	Sales (₹ in lakh) (Y)
Arithmetic mean	10	90
Standard deviation	3	12

Correlation coefficient between X and Y is 0.8

- Obtain the regression of Y on X
- What is the likely sales when the advertising budget is ₹ 15 lakh?

- Find y if the cost of living index is 200 for the following data:

Group	Food	Clothing	Fuel and Lighting	House Rent	Miscellaneous
I	180	120	160	300	200
W	4	5	3	y	2

(B) Attempt any TWO of the following questions (4 marks each): (8)

- A bill of ₹ 5,475 drawn on 19th January 2015 for 8 months was discounted on 28th February 2015 at 8% p.a. interest. What is the banker's discount? What is the cash value of the bill?

- Following table shows that all India Infant Mortality Rates (per '000) for years 1980 to 1986:

Years	1980	1981	1982	1983	1984	1985	1986
IMR	10	6	5	3	3	1	0

Fit a trend line to the above data by the method of least squares.

- Four new machines M_1, M_2, M_3 and M_4 are to be installed in a machine shop. There are five vacant places A, B, C, D and E available. Because of limited space, machine M_2 cannot be placed at C and M_3 cannot be placed at A. The cost matrix is given below:

Machines	Places				
	A	B	C	D	E
M_1	4	6	10	5	6
M_2	7	4	—	5	4
M_3	—	6	9	6	2
M_4	9	3	7	2	3

Find the optimal assignment schedule.

Q.6. (A) Attempt any TWO of the following questions (3 marks each):**(6) [14]**

- i. Obtain 4 yearly centered moving averages for the following time series:

Years	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991
Number of crime ('000)	40	42	43	42	44	44	43	46	47	45	46

- ii. Calculate Walsh's Price Index Number for the following data:

Commodity	Base Year		Current Year	
	Price	Quantity	Price	Quantity
L	4	8	3	2
M	6	16	8	9
N	8	18	7	32

- iii. A pair of dice is thrown 3 times. If getting a doublet is considered a success, find the probability of two successes.

(B) Attempt any ONE of the following questions (4 marks each):**(4)**

- i. The equations of two regression lines are $8x - 10y + 66 = 0$ and $40x - 18y = 214$. Find

- (a) The mean values of X and Y
(b) Correlation coefficient between X and Y

- ii. Maximize : $z = 60x + 50y$

Subject to : $x + 2y \leq 40$

$$3x + 2y \leq 60$$

$$x \geq 0, y \geq 0$$

(C) Attempt any ONE of the following questions (Activity) (4 marks each):**(4)**

- i. Find the sequence that minimizes the total elapsed time to complete the following jobs in the order A B. Find the total elapsed time and idle times for both the machines.

Jobs	I	II	III	IV	V	VI	VII
Machine A	7	16	19	10	14	15	5
Machine B	12	14	14	10	16	5	7

Solution:

Using the optimal sequence algorithm, the following optimal sequence can be obtained.

		IV	V	III		
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Total elapsed time is obtained as follows

Job	Machine A		Machine B	
	Time in	Time out	Time in	Time out
	0	5	5	12
	5	12	12	24
IV	12	22	24	34
V	22	36	36	52
III	36	55	55	69
	55	71	71	85
	71	86	86	91

∴ Total elapsed time T = 91 units

Idle time for machine A = units

Idle time for machine B = units

- ii. The probability distribution of X is as follows:

x	0	1	2	3	4
$P[X = x]$	0.1	k	$2k$	$2k$	k

Find (a) k (b) $P(X > 2)$ (c) $P(1 < X \leq 4)$

Solution:

The table gives a probability distribution and therefore

$$\sum P(X = x) = 1$$

$$P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) = 1$$

$$0.1 + k + 2k + 2k + k = 1$$

$$6k = 1 - 0.1$$

$$6k = 0.9$$

$$k = \boxed{}$$

$$P(X > 2) = P(X = 3) + P(X = 4)$$

$$= 2k + k$$

$$= \boxed{}$$

$$P(1 < X \leq 4) = P(X = \boxed{}) + P(X = 3) + P(X = 4)$$

$$= \boxed{} + 2k + k$$

$$= 0.75$$